

# **MULTIMODAL AI FRAMEWORK FOR PERSONALIZED AND CULTURALLY-AWARE SKIN DISEASE DIAGNOSIS, MONITORING, AND TREATMENT SUPPORT**

Project ID – 25-26J-222

Project Proposal Report

Bachelor of Science (Hons) Degree in Information Technology Specializing in  
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka(SLIIT)

August 2025

# **MULTIMODAL AI FRAMEWORK FOR PERSONALIZED AND CULTURALLY-AWARE SKIN DISEASE DIAGNOSIS, MONITORING, AND TREATMENT SUPPORT**

Project ID – 25-26J-222

Project Proposal Report

Bachelor of Science (Hons) Degree in Information Technology Specializing in  
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka(SLIIT)

August 2025

## ABSTRACT

Skin diseases are among the most common health conditions worldwide, affecting people of all ages and significantly impacting both physical health and psychological well-being. Early and accurate diagnosis is essential to prevent complications and ensure effective treatment. However, traditional diagnostic methods rely heavily on dermatologists, and access to such specialists is limited in many regions, including Sri Lanka. Although recent advancements in artificial intelligence (AI) have introduced automated skin disease classification systems, most existing approaches depend only on image-based analysis, which often leads to reduced accuracy when different diseases share similar visual characteristics.

To address these limitations, this research proposes a Multimodal AI Framework for Personalized and Culturally-Aware Skin Disease Diagnosis, Monitoring, and Treatment Support. The proposed system integrates multiple data sources, including skin images, patient-reported symptoms (text and audio), and clinical information, to provide a more comprehensive and context-aware diagnosis. Image data are processed using Convolutional Neural Networks (CNNs), while symptom data are analyzed using Natural Language Processing (NLP) and Automatic Speech Recognition (ASR) techniques. These modalities are combined using a fusion strategy to improve prediction accuracy.

In addition, the framework includes modules for disease progression monitoring, cross-cultural classification using domain adaptation techniques to reduce bias across different skin tones, and a personalized treatment recommendation system powered by knowledge graphs and Large Language Models (LLMs). The system also emphasizes explainability and fairness to ensure reliable and inclusive healthcare support.

The expected outcome of this research is a highly accurate, user-friendly, and scalable AI-based dermatology support system that enhances diagnostic performance, supports early detection, and provides personalized treatment recommendations. This system is designed as a decision-support tool to assist both patients and healthcare professionals, rather than replace medical expertise.

## **ACKNOWLEDGEMENT**

I would like to express my deepest gratitude to all those who supported me in the preparation and completion of this research proposal.

First and foremost, I extend my sincere appreciation to my supervisor, Mrs. Lokesha Weerasinghe, for her invaluable guidance, encouragement, and continuous support throughout this process. Her expert knowledge, constructive feedback, and patient mentorship have been instrumental in shaping the direction and quality of my research.

I am also profoundly grateful to my co-supervisor, Ms. Akshi De Silva, for her insightful advice, technical expertise, and thoughtful recommendations, which have greatly contributed to strengthening this proposal.

My heartfelt thanks further go to the CDAP team and the Faculty of Computing for providing essential resources, technical assistance, and an encouraging academic environment. Their support has played a vital role in the successful preparation of this proposal.

Finally, I extend my appreciation to my colleagues, peers, friends, and family for their constant motivation and encouragement throughout this journey. Their support has been a great source of strength and inspiration.

## TABLE OF CONTENTS

|                                           |      |
|-------------------------------------------|------|
| Declaration .....                         | iii  |
| Abstract .....                            | iv   |
| Acknowledgement .....                     | v    |
| Table Of Contents .....                   | vi   |
| List of Figures .....                     | vii  |
| List of Appendices .....                  | viii |
| List Of Abbreviations.....                | ix   |
| 1 Introduction .....                      | 10   |
| 1.1 Background.....                       | 11   |
| 1.2 Literature Review .....               | 13   |
| 1.3 Research Gap .....                    | 15   |
| 1.3.1 Related Work.....                   | 17   |
| 1.4 Research Problem.....                 | 19   |
| 1.5 Related Solution .....                | 20   |
| 2 Objectives .....                        | 21   |
| 2.1 Main Objectives .....                 | 21   |
| 2.2 Specific Objectives.....              | 22   |
| 3 Methodology .....                       | 23   |
| 3.1 Data Collection Method .....          | 24   |
| 3.2 High Level Architecture .....         | 25   |
| 3.3 Tools and Technologies .....          | 26   |
| 3.4 System Integration and Frontend ..... | 27   |
| 3.5 Software Solution .....               | 28   |
| 3.6 System Overview.....                  | 29   |
| 4 Commercialization.....                  | 30   |
| 5 Work Breakdown Structure.....           | 31   |
| 6 Gantt Chart.....                        | 32   |
| 7 Project Requiremnets .....              | 33   |
| 7.1 Functional Requirements.....          | 33   |
| 7.2 Nonfunctional Requirements .....      | 34   |
| 7.3 User Requirements .....               | 35   |

|                                                |    |
|------------------------------------------------|----|
| 8 Description of Personal and Facilities ..... | 36 |
| 9 Conclusion .....                             | 37 |
| 10 Budget and Budget Justification .....       | 38 |
| 11 References .....                            | 39 |

## **LIST OF FIGURES**

|                                                                     |    |
|---------------------------------------------------------------------|----|
| Figure:1 Comparison of skin disease classification approaches ..... | 16 |
| Figure:2 System High Level Architecture .....                       | 25 |
| Figure:3 Agile Software Development Cycle .....                     | 28 |
| Figure:4 System Overview .....                                      | 29 |
| Figure:5 Work Breakdown Structure .....                             | 31 |
| Figure:6 Gantt chart .....                                          | 32 |

## **LIST OF TABLES**

|                                                                            |    |
|----------------------------------------------------------------------------|----|
| Table 1 Feature Comparison of Existing Research and Proposed Project ..... | 18 |
| Table 2 Budget and Budget Justification .....                              | 38 |

## LIST OF ABBREVIATIONS

| Abbreviation | Description                                                                     |
|--------------|---------------------------------------------------------------------------------|
| AI           | Artificial Intelligence                                                         |
| CNN          | Convolutional Neural Network                                                    |
| RNN          | Recurrent Neural Network                                                        |
| NLP          | Natural Language Processing                                                     |
| ML           | Machine Learning                                                                |
| DL           | Deep Learning                                                                   |
| MFCC         | Mel-Frequency Cepstral Coefficients (commonly used in audio feature extraction) |
| API          | Application Programming Interface                                               |
| GPU          | Graphics Processing Unit                                                        |
| ACC          | Accuracy (performance metric)                                                   |
| AUC          | Area Under the Curve (evaluation metric)                                        |



# 1 INTRODUCTION

Skin diseases are among the most common health conditions worldwide, affecting millions of people across different age groups, ethnicities, and geographic regions. These conditions range from mild infections to severe chronic and life-threatening diseases such as melanoma. According to the World Health Organization, skin diseases contribute significantly to the global burden of disease, impacting both physical health and psychological well-being [1]. Early diagnosis and timely treatment are essential to reduce complications and improve patient outcomes.

Traditionally, the diagnosis of skin diseases relies on dermatologists' expertise and clinical examination. However, access to dermatological care is limited in many regions, especially in developing countries such as Sri Lanka. This limitation often leads to delayed diagnosis, misdiagnosis, and inadequate treatment. In recent years, Artificial Intelligence (AI), particularly deep learning, has shown promising results in medical image analysis and disease classification. Convolutional Neural Networks (CNNs) have demonstrated dermatologist-level performance in identifying skin diseases from images [2], [5]. Large datasets such as ISIC and HAM10000 have further supported the development of automated skin disease classification systems [3], [4].

Despite these advancements, most existing AI-based systems rely only on image data. This creates significant limitations because many skin diseases share similar visual characteristics, making accurate diagnosis difficult without additional context. Dermatologists typically consider not only visual features but also patient-reported symptoms such as itching, pain, or burning sensations. Ignoring these symptoms reduces the effectiveness and reliability of automated systems. Therefore, there is a growing need for more advanced approaches that incorporate multiple types of data to improve diagnostic accuracy.

Multimodal learning has emerged as a powerful solution to address these limitations by combining different data sources such as images, text, and audio. Recent research shows that integrating clinical data with image-based models improves performance and provides more context-aware predictions [6]. Natural Language Processing (NLP) models like BERT enable effective analysis of patient symptoms in textual form [9], while speech recognition technologies allow the use of voice-based symptom descriptions [10]. These advancements support the development of more comprehensive and patient-centered diagnostic systems.

Another critical challenge in dermatological AI is the issue of bias and fairness. Many datasets used for training AI models are imbalanced and overrepresent lighter skin tones, resulting in reduced accuracy for darker skin tones and underrepresented populations [13], [14]. This creates disparities in healthcare outcomes and limits the generalizability of AI systems. Domain adaptation techniques, such as adversarial learning, have been proposed to address this problem by enabling models to perform consistently across different data distributions [11], [12]. Ensuring fairness and inclusivity is essential for building reliable AI systems that can be used globally.

In addition to diagnosis, there is an increasing demand for systems that can provide personalized treatment recommendations. Traditional systems often fail to consider individual factors such as age, skin type, allergies, and cultural practices. Recent advancements in knowledge graphs and Large Language Models (LLMs) enable the development of intelligent systems that can generate personalized, explainable, and context-aware treatment suggestions [17]. These technologies help bridge the gap between diagnosis and real-world clinical decision-making.

Therefore, this research proposes a Multimodal AI Framework for Personalized and Culturally-Aware Skin Disease Diagnosis, Monitoring, and Treatment Support. The proposed system integrates image data, patient symptoms, and clinical information to improve diagnostic accuracy, reduce bias, and provide personalized recommendations. By combining multiple AI techniques, including deep learning, NLP, speech recognition, and domain adaptation, this framework aims to create a more accurate, fair, and user-friendly dermatological support system.

## 1.1 Background

Skin diseases are one of the most widespread non-communicable health conditions globally, affecting millions of individuals and contributing significantly to the overall disease burden [1]. Common conditions such as eczema, acne, psoriasis, and fungal infections not only affect physical health but also have a strong impact on psychological well-being, including stress, low self-esteem, and social stigma. In severe cases, such as skin cancer, delayed diagnosis can lead to life-threatening consequences. Therefore, early detection and continuous monitoring are critical for effective treatment and improved patient outcomes.

However, access to dermatological care remains limited, especially in developing countries like Sri Lanka. The shortage of dermatologists, high consultation costs, and long waiting times make it difficult for patients to receive timely diagnosis and treatment. As a result, many individuals rely on self-diagnosis or non-medical advice, which can worsen the condition. These challenges highlight the need for accessible, affordable, and efficient healthcare solutions.

With the advancement of Artificial Intelligence (AI), particularly deep learning, new opportunities have emerged in the field of dermatology. Convolutional Neural Networks (CNNs) have shown strong performance in image-based skin disease classification by learning complex visual patterns such as texture, color, and shape [2], [7]. Publicly available datasets like ISIC and HAM10000 have further accelerated research in this area by providing large collections of labeled skin images [3], [4]. These developments have enabled AI systems to achieve performance levels comparable to dermatologists in certain classification tasks [5].

Despite these improvements, most existing AI-based systems are limited to image-only analysis. In real clinical practice, dermatologists consider additional factors such as patient-reported symptoms (e.g., itching, pain, burning), medical history, and environmental conditions when making a diagnosis. Ignoring these contextual factors reduces the accuracy and reliability of automated systems, especially when diseases have similar visual appearances.

To overcome this limitation, researchers have started exploring **multimodal learning**, which combines multiple types of data such as images, text, and audio to improve diagnostic performance [6]. Natural Language Processing (NLP) techniques, including models like BERT, allow the analysis of symptom descriptions provided by patients [9]. Similarly, speech

recognition technologies enable the use of voice-based inputs, which is especially useful in low-literacy populations [10]. By integrating these different data sources, multimodal systems can provide more accurate and context-aware predictions.

Another important challenge in dermatological AI is the issue of bias and fairness. Many existing datasets are imbalanced and predominantly represent lighter skin tones, which leads to reduced model performance for darker skin tones and underrepresented populations [13], [14]. This creates disparities in diagnosis and limits the global applicability of AI systems. Domain adaptation techniques, such as adversarial learning, have been introduced to address this issue by improving model generalization across different data distributions [11], [12].

Furthermore, traditional AI systems mainly focus on disease detection and do not provide personalized treatment recommendations. However, effective treatment depends on multiple patient-specific factors, including age, skin type, allergies, and cultural practices. Recent advancements in knowledge graphs and Large Language Models (LLMs) enable the development of intelligent systems that can generate personalized, explainable, and context-aware treatment suggestions [17].

Therefore, there is a clear need for a comprehensive system that not only improves diagnostic accuracy but also ensures fairness, personalization, and real-world usability. This research addresses these challenges by proposing a multimodal AI framework that integrates multiple data sources and advanced AI techniques to support skin disease diagnosis, monitoring, and treatment.

## 1.2 Literature Review

The application of Artificial Intelligence (AI) in dermatology has rapidly increased over the past decade, particularly in the areas of skin disease detection and classification. Early research mainly focused on image-based analysis using deep learning techniques, especially Convolutional Neural Networks (CNNs). Esteva et al. demonstrated that CNN models can achieve dermatologist-level accuracy in classifying skin cancer from clinical images [2]. Similarly, Brinker et al. showed that deep neural networks can outperform dermatologists in certain melanoma classification tasks [5]. These studies highlight the strong potential of AI in improving diagnostic accuracy and supporting clinical decision-making.

Publicly available datasets such as ISIC and HAM10000 have played a major role in advancing research in this domain [3], [4]. These datasets provide large collections of labeled skin images, enabling researchers to train and evaluate machine learning models effectively. Various architectures, including ResNet, DenseNet, and MobileNet, have been successfully applied to skin disease classification tasks, achieving high levels of accuracy and efficiency [7]. In addition, transfer learning techniques have been widely used to improve model performance, especially when dealing with limited medical datasets.

Despite the success of image-based models, several limitations have been identified. One major issue is that many skin diseases share similar visual features, making it difficult for models to differentiate between conditions using only images. In real-world clinical settings, dermatologists rely not only on visual examination but also on patient-reported symptoms such as itching, pain, and burning sensations. Therefore, researchers have started exploring multimodal approaches that combine images with additional data sources to improve diagnostic performance [6].

Natural Language Processing (NLP) has been introduced to analyze textual symptom descriptions provided by patients. Models such as BERT have shown strong performance in understanding and

extracting meaningful information from clinical text data [9]. In addition, speech recognition technologies have been used to convert voice-based symptom descriptions into text, enabling the integration of audio data into diagnostic systems [10]. These advancements support the development of more comprehensive and patient-centered AI systems.

Another important area of research is multimodal learning, which integrates multiple data types such as images, text, and metadata. Studies have shown that combining these modalities leads to improved accuracy and robustness compared to unimodal systems [6]. Multimodal models are particularly effective in handling complex medical scenarios where visual information alone is insufficient.

However, a significant challenge in dermatological AI is the issue of bias and fairness. Many existing models are trained on datasets that predominantly represent lighter skin tones, resulting in reduced performance for darker skin tones and underrepresented populations [13], [14]. This bias can lead to inaccurate predictions and unequal healthcare outcomes. To address this problem, domain adaptation techniques have been proposed. Ganin et al. introduced adversarial domain adaptation methods to learn domain-invariant features, improving model generalization across different datasets [11]. Similarly, Tzeng et al. proposed adversarial discriminative domain adaptation to enhance transferability between domains [12]. These techniques help reduce bias and improve fairness in AI-based systems.

In addition to diagnosis, recent research has focused on personalized treatment recommendations. Traditional systems often provide generic suggestions without considering patient-specific factors. Knowledge graphs have been used to represent relationships between diseases, treatments, and patient attributes, enabling more structured and meaningful analysis [17]. When combined with Large Language Models (LLMs), these systems can generate personalized, explainable, and context-aware treatment recommendations, improving patient satisfaction and safety.

Overall, the literature shows that while significant progress has been made in AI-based skin disease diagnosis, there are still important gaps. Existing systems lack multimodal integration, fairness across diverse populations, and personalized treatment capabilities. This research aims to address these limitations by developing a comprehensive multimodal AI framework that integrates multiple data sources and advanced techniques to improve accuracy, fairness, and usability in dermatological care.

### 1.3 Research Gap

Although significant progress has been made in applying Artificial Intelligence (AI) for skin disease diagnosis, several important limitations still exist in current research and systems. These gaps reduce the effectiveness, fairness, and real-world applicability of existing solutions.

Firstly, most existing AI-based dermatology systems rely only on **image-based classification**. While Convolutional Neural Networks (CNNs) have shown high accuracy in recognizing skin diseases, they depend solely on visual features such as color, texture, and shape. However, many skin diseases share similar visual characteristics, making it difficult to achieve accurate diagnosis using images alone. In real clinical practice, dermatologists also consider patient-reported symptoms such as itching, pain, and burning sensations. The lack of integration of these symptoms creates a major gap in current systems.

Secondly, there is limited use of **multimodal data integration**. Although some studies have explored combining images with textual data, the integration of **audio-based symptom descriptions** is still underexplored. Voice input is a natural and accessible way for patients to describe their symptoms, especially in low-literacy populations. The absence of audio-based analysis in most systems limits their usability and inclusiveness.

Another critical gap is the lack of **fairness and generalization across different skin tones and cultural contexts**. Many datasets used in training AI models are imbalanced and heavily represent lighter skin tones. As a result, models often perform poorly on darker skin tones, leading to biased predictions and unequal healthcare outcomes. While domain adaptation techniques have been proposed, their application in dermatology is still limited and requires further improvement.

Furthermore, most existing systems focus only on **disease detection and classification**, without supporting **disease progression monitoring**. Skin diseases often change over time, and tracking these changes is important for evaluating treatment effectiveness. The absence of temporal analysis in many systems reduces their clinical usefulness.

In addition, there is a lack of **personalized treatment recommendation systems**. Current AI models typically provide only diagnostic outputs without considering patient-specific factors such as age, skin type, allergies, and cultural practices. This limits their ability to provide meaningful and practical healthcare support. Although knowledge graphs and Large Language Models (LLMs) have shown potential in generating personalized recommendations, their integration into dermatological systems is still not widely explored.

Finally, many AI systems lack **explainability and transparency**, which are essential in medical applications. Users and healthcare professionals need to understand how predictions are made in order to trust and effectively use these systems.

### 1.3.1 Related Work

Several studies have explored the use of Artificial Intelligence (AI) in skin disease diagnosis, mainly focusing on image-based classification. One of the most influential works by Esteva et al. demonstrated that deep Convolutional Neural Networks (CNNs) can achieve performance comparable to dermatologists in classifying skin cancer using clinical images [2]. Similarly, Brinker et al. showed that deep learning models can outperform human experts in melanoma detection tasks [5]. These studies highlight the strong capability of CNN-based models in medical image analysis.

In addition, large-scale datasets such as ISIC and HAM10000 have significantly contributed to the development of skin disease classification systems [3], [4]. These datasets provide diverse and labeled images, allowing researchers to train deep learning models effectively. Various architectures such as ResNet, DenseNet, and MobileNet have been applied in this domain, achieving high classification accuracy [7]. Transfer learning has also been widely used to improve performance when training data is limited.

Despite these advancements, many studies have identified limitations in image-only approaches. To overcome this, some researchers have explored integrating additional data sources. Liu et al. compared AI systems with healthcare professionals and showed that combining different types of medical data can improve diagnostic performance [6]. Similarly, recent studies have introduced multimodal approaches that combine image data with clinical metadata or textual symptom descriptions to enhance model accuracy and robustness.

Natural Language Processing (NLP) techniques have also been applied in healthcare to analyze patient-reported symptoms. Models such as BERT have shown strong performance in understanding clinical text and extracting meaningful information [9]. Additionally, speech recognition technologies have been used to convert voice-based inputs into text, enabling the use of audio data in medical systems [10]. However, the integration of audio-based symptom descriptions in dermatology remains limited.

Another important area of related work is addressing bias and fairness in AI models. Studies have shown that many dermatology datasets are imbalanced, leading to reduced performance for darker skin tones [13], [14]. To address this issue, domain adaptation techniques have been introduced. Ganin et al. proposed adversarial domain adaptation to learn domain-invariant features [11], while Tzeng et al. developed methods to improve transferability between different datasets [12]. These approaches help improve generalization and reduce bias in classification models.

Furthermore, recent research has explored the use of knowledge graphs and Large Language Models (LLMs) for personalized healthcare. Knowledge graphs allow structured representation of relationships between diseases, treatments, and patient characteristics, while LLMs can generate human-like explanations and recommendations [17]. These technologies enable the development of personalized and explainable treatment systems, although their application in dermatology is still emerging.

## 1.4 Research Problem

Skin diseases are widely prevalent and require accurate diagnosis and proper treatment to prevent complications. However, the current process of diagnosing skin diseases faces several challenges, especially in regions with limited access to dermatologists. Traditional diagnosis depends heavily on expert knowledge, making it time-consuming, costly, and sometimes inaccessible to many patients.

Although Artificial Intelligence (AI) has been introduced to support skin disease diagnosis, most existing systems rely primarily on **image-based classification**. These systems analyze visual features such as color, texture, and shape using deep learning models. While they have shown high accuracy in controlled environments, they often struggle in real-world scenarios where different skin diseases share similar visual patterns. As a result, relying only on image data can lead to incorrect or uncertain predictions.

Another major issue is the lack of **contextual information** in current systems. In clinical practice, dermatologists consider patient-reported symptoms such as itching, pain, and burning sensations, along with medical history and environmental factors. However, most AI-based systems ignore these important factors, reducing their diagnostic reliability and effectiveness.

In addition, many existing models suffer from **bias and poor generalization across different skin tones**. Most training datasets are imbalanced and predominantly represent lighter skin tones, which leads to reduced accuracy for darker skin tones and underrepresented populations. This creates unfair and unreliable outcomes, limiting the usability of these systems in diverse populations.

Furthermore, current systems focus mainly on disease classification and do not support **disease progression monitoring**. Skin conditions often change over time, and monitoring these changes is essential for evaluating treatment effectiveness. The absence of temporal analysis limits the practical use of AI systems in continuous healthcare.

Another important limitation is the lack of **personalized treatment recommendations**. Most AI systems provide only diagnostic results without considering patient-specific factors such as age, skin type, allergies, and cultural practices. This makes it difficult to provide meaningful and actionable healthcare support.

Finally, many AI models lack **explainability and transparency**, which are critical in medical applications. Users and healthcare professionals need to understand how predictions are made in order to trust and effectively use these systems.

## 1.5 Research Solution

To address the identified challenges in existing skin disease diagnosis systems, this research proposes a **Multimodal AI Framework for Personalized and Culturally-Aware Skin Disease Diagnosis, Monitoring, and Treatment Support**. The proposed solution aims to improve diagnostic accuracy, ensure fairness across diverse populations, and provide personalized healthcare support by integrating multiple data sources and advanced AI techniques.

Firstly, the system introduces a **multimodal approach** that combines skin images with patient-reported symptoms in both text and audio formats. Image data will be processed using Convolutional Neural Networks (CNNs) to extract visual features such as texture, color, and lesion patterns. At the same time, symptom data will be analyzed using Natural Language Processing (NLP) models and Automatic Speech Recognition (ASR) to convert voice inputs into meaningful textual information. These different data types will then be combined using a fusion strategy to generate a more accurate and context-aware diagnosis.

Secondly, the framework incorporates a **symptom-aware disease classification module**, which enhances traditional image-based models by including patient symptoms in the prediction process. This helps the system differentiate between diseases with similar visual characteristics and improves overall diagnostic reliability.

To address the issue of bias and ensure fairness, the proposed system integrates **domain adaptation techniques**. These methods enable the model to learn domain-invariant features, allowing it to perform consistently across different skin tones, ethnic groups, and imaging conditions. This ensures that the system provides equitable and accurate predictions for all users.

In addition, the framework includes a **disease progression monitoring module**, which analyzes sequential images over time to track changes in skin conditions. This feature helps in evaluating treatment effectiveness and supports continuous patient care, making the system more useful in real-world clinical scenarios.

Another key component of the solution is the **personalized treatment recommendation module**. This module uses knowledge graphs to represent relationships between diseases, treatments, patient characteristics, and potential risks such as allergies. A Large Language Model (LLM) is then used to generate personalized, explainable, and context-aware treatment suggestions based on individual patient data. This ensures that recommendations are tailored to each user's needs and cultural context.

Furthermore, the proposed system emphasizes **explainability and transparency** by providing clear reasoning for its predictions and recommendations. This helps build trust among users and healthcare professionals, which is essential for the adoption of AI systems in medical applications.



## 2 OBJECTIVES

### 2.1 Main Objectives

The main objective of this research is to design and develop a **Multimodal AI Framework for Personalized and Culturally-Aware Skin Disease Diagnosis, Monitoring, and Treatment Support** that can significantly improve the accuracy, reliability, and usability of existing dermatological AI systems.

This system aims to overcome the limitations of traditional image-based diagnosis by integrating multiple types of data, including skin images, patient-reported symptoms (text and audio), and clinical information. In addition, the framework focuses on reducing bias across different skin tones and cultural backgrounds, ensuring fair and inclusive healthcare support for diverse populations.

Furthermore, the proposed system intends to provide not only accurate disease classification but also continuous monitoring of disease progression and personalized treatment recommendations. By incorporating explainable AI techniques, the system will generate transparent and understandable outputs, making it more trustworthy and useful for both patients and healthcare professionals. Overall, this research aims to develop a comprehensive, intelligent, and user-friendly dermatology support system that can be applied in real-world healthcare environments, especially in regions with limited access to dermatologists.

### 2.2 Specific Objectives

To achieve the main objective, the following specific objectives are defined:

1. **To develop an accurate image-based skin disease classification model**  
This objective focuses on building a deep learning model using Convolutional Neural Networks (CNNs) to analyze skin images. The model will extract important visual features such as texture, color, and lesion patterns to identify different types of skin diseases with high accuracy.
2. **To implement symptom-based analysis using text and audio data**  
This objective aims to capture patient-reported symptoms in both textual and voice formats. Natural Language Processing (NLP) techniques will be used to analyze textual data, while Automatic Speech Recognition (ASR) will convert audio inputs into text. This allows the system to better understand the patient's condition beyond visual information.
3. **To design and implement a multimodal data fusion mechanism**  
This objective focuses on combining multiple data sources, including images and

symptoms, to improve diagnostic performance. A fusion strategy will be used to integrate outputs from different models and generate a more accurate and context-aware prediction.

**4. To improve fairness and reduce bias in the system**

This objective addresses the issue of dataset imbalance and bias toward certain skin tones. Domain adaptation techniques will be applied to ensure that the model performs consistently across different skin tones, ethnic groups, and environmental conditions, promoting fairness and inclusivity.

**5. To develop a disease progression monitoring module**

This objective involves analyzing sequential images over time to track changes in skin conditions. The system will help evaluate whether a disease is improving, worsening, or remaining stable, which is important for effective treatment and long-term care.

**6. To build a personalized treatment recommendation system**

This objective aims to provide tailored treatment suggestions based on individual patient characteristics such as age, skin type, allergies, and medical history. Knowledge graphs will be used to represent relationships between diseases and treatments, while Large Language Models (LLMs) will generate clear and user-friendly recommendations.

**7. To ensure explainability and transparency of AI predictions**

This objective focuses on making the system's outputs understandable to users. The system will provide explanations for its predictions and recommendations, helping users and healthcare professionals trust and effectively use the system.

**8. To design and develop a user-friendly interface for the system**

This objective ensures that the system is easy to use for both patients and healthcare providers. The interface will allow users to upload images, input symptoms, and receive results in a clear and organized manner.

### 3 METHODOLOGY

The proposed research follows a structured methodology to develop a **Multimodal AI Framework for Skin Disease Diagnosis, Monitoring, and Treatment Support**. The system integrates multiple data sources, including images, text, and audio, to improve diagnostic accuracy and provide personalized healthcare solutions. The methodology consists of several key stages, including data collection, preprocessing, model development, multimodal fusion, and system implementation.

#### 3.1 Data Collection Method

The first step of the methodology involves collecting relevant data required for training and testing the models. The system will use multiple types of data:

- **Skin Image Data:** Public datasets such as ISIC and HAM10000 will be used to obtain labeled skin disease images. These datasets include various skin conditions with different characteristics.
- **Symptom Data (Text):** Patient-reported symptoms such as itching, pain, and burning will be collected or simulated in textual format.
- **Audio Data:** Voice recordings describing symptoms will be collected and converted into text using speech recognition techniques.
- **Clinical Data:** Additional information such as patient age, skin type, and medical history may also be included to enhance system performance.

This multimodal dataset allows the system to learn from both visual and contextual information.

#### 3.2 Data Preprocessing

Before feeding data into the models, preprocessing is performed to improve data quality and consistency.

- **Image Preprocessing:**
  - Resize images to a fixed size
  - Normalize pixel values
  - Apply techniques such as noise reduction and contrast enhancement (e.g., CLAHE)
  - Perform data augmentation (rotation, flipping) to improve model generalization
- **Text Preprocessing:**
  - Remove unnecessary words and symbols
  - Tokenization and text cleaning
  - Convert text into numerical representations using embeddings
- **Audio Preprocessing:**
  - Convert speech to text using ASR
  - Extract audio features such as MFCC (if needed)

### 3.3 Tools and Technologies

The development of the proposed multimodal AI framework utilizes a combination of modern tools and technologies to support data processing, model development, system integration, and user interaction. These technologies are carefully selected to ensure efficiency, scalability, and accuracy throughout the system.

Python is used as the primary programming language for this research due to its simplicity, flexibility, and extensive support for machine learning and data science applications. It provides a wide range of libraries that facilitate image processing, natural language processing, and deep learning, making it highly suitable for developing AI-based systems.

For machine learning and deep learning tasks, frameworks such as TensorFlow and PyTorch are used to build and train models. These frameworks support the implementation of Convolutional Neural Networks (CNNs) for image classification and transformer-based models for text analysis. In addition, Scikit-learn is used for data preprocessing, performance evaluation, and basic machine learning operations such as data splitting and metric calculations.

Natural Language Processing (NLP) is implemented using the Hugging Face Transformers library, which enables the use of advanced models such as BERT and DistilBERT for analyzing patient-reported symptoms. These models help extract meaningful insights from textual data and improve the system's ability to understand user input. For audio processing, Automatic Speech Recognition (ASR) technologies are used to convert voice-based symptom descriptions into text, allowing the system to support voice input and improve accessibility.

Image preprocessing and enhancement are performed using OpenCV and Pillow (PIL), which provide functionalities such as resizing, filtering, and noise reduction. These tools help improve the quality of input images and enhance model performance. Data manipulation and analysis are carried out using libraries such as NumPy and Pandas, while visualization tools like Matplotlib and Seaborn are used to present model results and performance comparisons.

The development and testing of models are conducted using environments such as Jupyter Notebook and Google Colab, which provide interactive platforms for experimentation. Anaconda is used to manage dependencies and maintain a stable development environment.

For system implementation, frontend technologies such as React.js, HTML, CSS, and JavaScript are used to design a user-friendly interface. This interface allows users to upload images, input symptoms, and view results easily. The backend is developed using frameworks such as Flask, Django, or Node.js, which handle data processing, model integration, and communication between system components.

Databases such as MongoDB or MySQL are used to store user data, predictions, and system outputs securely. Version control is managed using Git and GitHub, which support collaborative development and code management. Additionally, REST APIs are used to connect the frontend and backend, ensuring smooth data communication within the system.

Overall, the combination of these tools and technologies enables the development of a robust and scalable multimodal AI system that supports efficient data processing, accurate model performance, and seamless user interaction.

### **3.5 Software solution**

The Software Development Life Cycle (SDLC) provides a structured framework for developing the Symptom-Aware Disease Classification system, guiding the team through design, implementation, testing, and deployment while ensuring high-quality, reliable software. Following a traditional SDLC helps in defining clear phases such as requirements analysis, system design, model development, integration, and evaluation, reducing risks and ensuring that the final system meets both clinical and user expectations.

However, rigid sequential models can limit flexibility when adjustments are needed for evolving datasets, patient feedback, or model improvements. To overcome these limitations, the development team adopts Agile methodologies, which enable iterative design, incremental model updates, and continuous testing. Among Agile frameworks, Scrum is employed for its focus on teamwork, iterative progress, and frequent stakeholder feedback. Through short sprints and regular reviews, developers can quickly respond to changes, integrate new audio or image processing features, and refine the user interface. This approach ensures that the software remains user-centered, culturally-aware, and technically robust, providing clinicians and patients with a reliable, real-time tool for symptom-aware skin disease classification.

### 3.6 System Overview

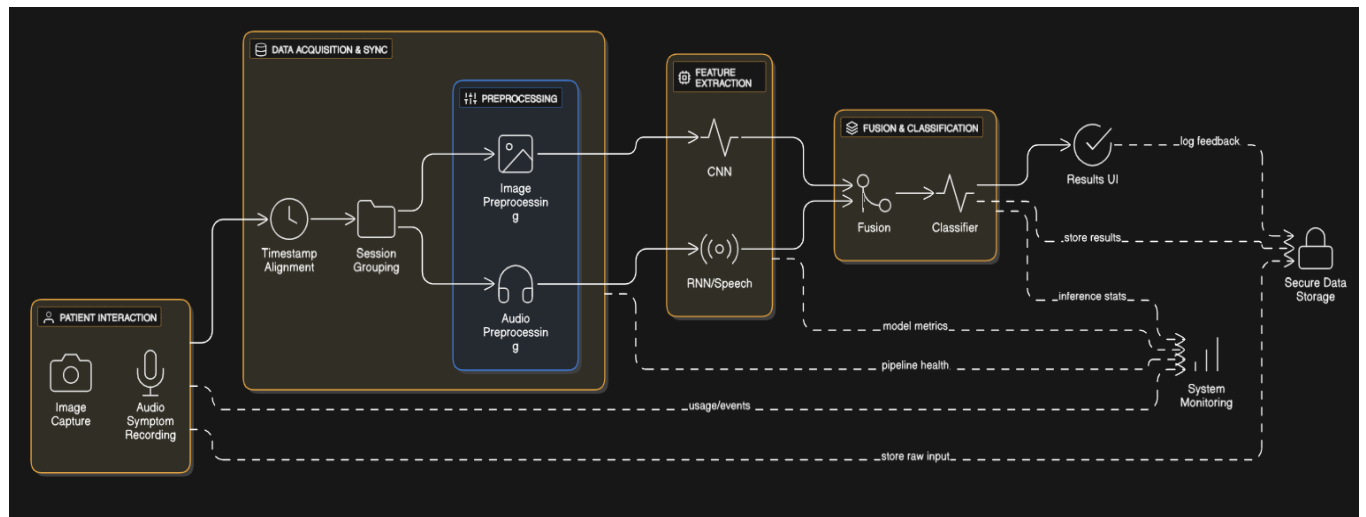


Figure 4: System Overview

## 4 COMMERCIALIZATION POTENTIAL

The Symptom-Aware Disease Classification system has strong potential for commercialization within healthcare and telemedicine markets. By integrating image-based dermatological analysis with patient-reported audio symptoms, the system delivers more accurate, personalized, and culturally-aware diagnoses. This capability makes it highly attractive for clinics, hospitals, and telehealth providers aiming to enhance patient care while reducing misdiagnosis rates.

One promising application is teledermatology services, where the system can be embedded into remote consultation platforms. This allows healthcare providers to deliver real-time skin disease diagnosis to patients in rural or underserved regions, improving access to specialized care. The ability to analyze both visual and audio symptom data in real time ensures more reliable consultations, even when in-person visits are not feasible.

Another commercial avenue lies in patient-centered healthcare applications. Mobile apps integrated with the system can facilitate continuous monitoring of chronic skin conditions, track symptom progression, and provide automated alerts if conditions worsen. This not only empowers patients with actionable insights but also encourages adherence to treatment plans, improving long-term health outcomes.

The system also has significant potential as a clinical decision support **tool** for hospitals and dermatology clinics. By supplementing traditional examinations, it can enhance diagnostic accuracy, reduce clinician workload, and optimize workflow efficiency. Additionally, anonymized and aggregated data generated by the system can provide valuable insights to pharmaceutical companies and healthcare organizations, supporting research and informing treatment strategies.

With cloud-based architecture, scalable deployment, and a culturally-inclusive design, the solution can be monetized through subscription-based telemedicine services, SaaS offerings for healthcare institutions, and integration into mobile health applications, positioning it for wide adoption and sustainable revenue generation.

## 5 WORK BREAKDOWN STRUCTURE

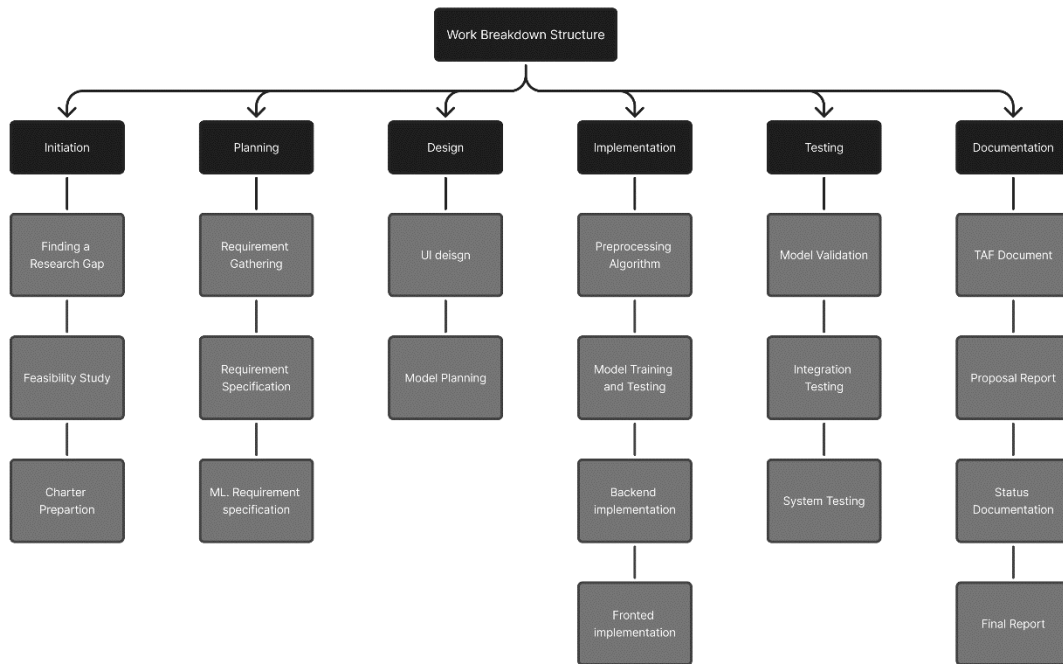


Figure 5: work breakdown structure



## 6 GANTT CHART

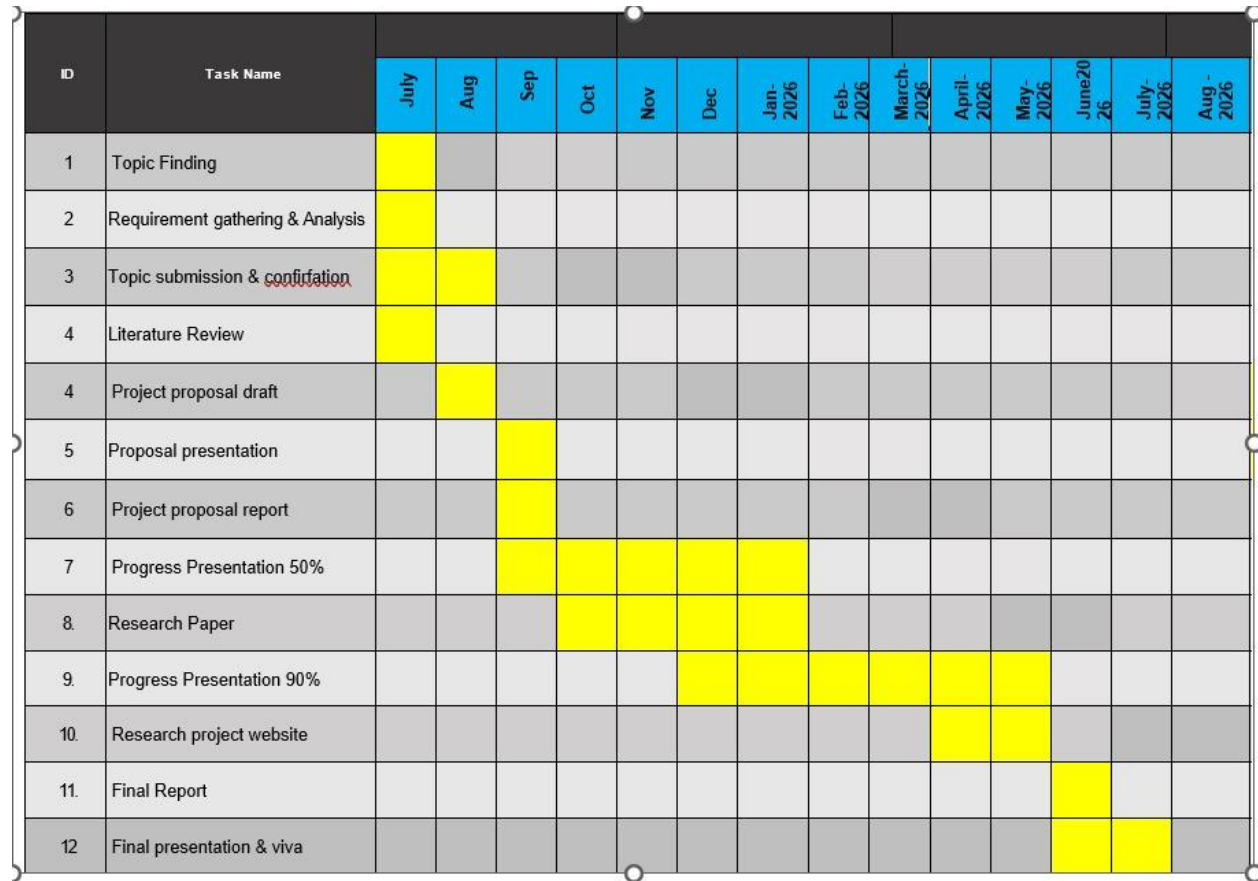


Figure 6: Gantt Chart

## 7 PROJECT REQUIREMENTS

### 7.1 Functional Requirements

- **Image Processing** – The system must accept dermatology images and perform preprocessing steps such as resizing, normalization, and augmentation to prepare inputs for CNN-based or Transformer-based classification models.
- **Audio Input Handling** – The system must allow patients to record and submit audio descriptions of symptoms (e.g., itching, pain, burning) in multiple languages such as Sinhala and Tamil.
- **Speech-to-Feature Extraction** – The system must process audio inputs and extract both acoustic and linguistic features, including tone, pitch, and phoneme patterns, for symptom analysis.
- **Text Processing (NLP)** – The system must process textual symptom descriptions using NLP techniques to understand patient-reported conditions.
- **Multimodal Fusion** – The system must integrate image-based features and symptom-based features (text/audio) using fusion techniques such as weighted fusion, concatenation, or attention mechanisms.
- **Classification & Diagnosis** – The system must classify skin diseases into predefined categories using combined multimodal data and provide accurate predictions.
- **Top-3 Prediction Support** – The system must generate multiple possible predictions (Top-3 results) to improve reliability in medical diagnosis.
- **Disease Progress Monitoring** – The system must allow tracking of skin conditions over time by comparing sequential images and identifying progression patterns.
- **Personalized Treatment Recommendation** – The system must provide treatment suggestions based on diagnosis, patient-specific factors, and medical knowledge.
- **User Interface (UI)** – The system must provide an easy-to-use interface for patients to upload images, record audio, and view results, and for dermatologists to review diagnostic outputs.
- **Data Management** – The system must securely store, manage, and retrieve patient data for analysis, prediction, and system improvement.
- **Feedback Generation** – The system must provide diagnostic results along with confidence scores and explanations to users or healthcare professionals.
- **User Authentication (Optional)** – The system may include login and authentication features to manage user accounts and protect data.

## 7.2 Nonfunctional Requirements

- **Accuracy** – The system must achieve high accuracy in disease classification using multimodal data.
- **Performance** – The system must provide fast response times for image processing, audio analysis, and prediction generation.
- **Scalability** – The system must be able to handle increasing numbers of users and large datasets without performance degradation.
- **Security & Privacy** – The system must ensure secure storage and transmission of sensitive patient data, following privacy standards.
- **Usability** – The system must provide a simple, clear, and user-friendly interface suitable for both patients and medical professionals.
- **Reliability** – The system must function consistently with minimal errors and downtime.
- **Robustness** – The system must handle low-quality images, noisy audio inputs, and incomplete data without significant performance loss.
- **Fairness** – The system must provide consistent performance across different skin tones, ethnic groups, and cultural backgrounds.
- **Maintainability** – The system must be easy to update, modify, and maintain over time.
- **Compatibility** – The system must be accessible across multiple platforms such as web and mobile devices.
- **Explainability** – The system must provide understandable explanations for predictions and recommendations to build user trust.
- **Resource Efficiency** – The system must efficiently utilize computational resources such as CPU and GPU during processing.

## **8 DESCRIPTION OF PERSONAL AND FACILITIES**

- Mrs Lokesha Weerasinghe – Sri Lanka Institute of Information Technology (SLIIT)
- Ms.Akshi De Silva- Sri Lanka Institute of Information Technology (SLIIT)

## 9 CONCLUSION

This research presents a **Multimodal AI Framework for Personalized and Culturally-Aware Skin Disease Diagnosis, Monitoring, and Treatment Support**, addressing several key limitations found in existing dermatology systems. Traditional AI-based approaches mainly rely on image-only analysis, which often leads to reduced accuracy when diseases share similar visual features and when important contextual information is missing.

The proposed system overcomes these challenges by integrating multiple data sources, including skin images, patient-reported symptoms in text and audio formats, and clinical information. By combining these modalities through a fusion mechanism, the system is able to provide more accurate and context-aware predictions. In addition, the inclusion of domain adaptation techniques helps reduce bias and ensures fair performance across different skin tones and populations.

Another significant contribution of this research is the inclusion of a disease progression monitoring module, which allows continuous tracking of skin conditions over time. This feature enhances the practical value of the system in real-world healthcare scenarios. Furthermore, the integration of knowledge graphs and Large Language Models (LLMs) enables the system to generate personalized and explainable treatment recommendations based on individual patient characteristics.

Overall, the proposed framework not only improves diagnostic accuracy but also enhances usability, fairness, and personalization. It is designed as a **decision-support tool** to assist both patients and healthcare professionals, rather than replace medical expertise. The system is especially beneficial in regions with limited access to dermatologists, providing an accessible and scalable healthcare solution.

In conclusion, this research contributes to the advancement of AI in dermatology by developing a comprehensive, intelligent, and user-centered system. Future improvements can include the use of larger and more diverse datasets, real-world clinical validation, and further enhancement of model performance and explainability to ensure wider adoption and impact.

## 10 BUDGET AND BUDGET JUSTIFICATION

| Item                            | Description                                                                                                               | Cost Estimation |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------|
| Hardware & Cloud Infrastructure | Use of a high-performance GPU or cloud computing credits for model training and real-time processing of images and audio. | 100 000 LKR     |
| Software & Tools                | Free or low-cost Python libraries, open-source AI frameworks, and basic annotation tools.                                 | 24 000 LKR      |
| Data Acquisition & Annotation   | Use of publicly available datasets and minimal manual annotation for audio symptom integration.                           | 150,000 LKR     |
| Miscellaneous Expenses          | Minor expenses such as software updates, minor hardware needs, or internet/data costs.                                    | 50 000 LKR      |

Table 2 Budget and Budget Justification

## 11 REFERENCES

1. World Health Organization, *Global Health Estimates: Skin Diseases and Conditions*, WHO, 2023.
2. A. Esteva, B. Kuprel, R. A. Novoa, et al., “Dermatologist-level classification of skin cancer with deep neural networks,” *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.
3. N. Codella, D. Gutman, M. Celebi, et al., “Skin lesion analysis toward melanoma detection: A challenge at the International Symposium on Biomedical Imaging,” *IEEE ISBI*, pp. 168–172, 2018.
4. P. Tschandl, C. Rosendahl, H. Kittler, “The HAM10000 dataset: A large collection of multi-source dermatoscopic images of common pigmented skin lesions,” *Scientific Data*, vol. 5, 2018.
5. J. Brinker, A. Hekler, S. Enk, et al., “Deep neural networks are superior to dermatologists in melanoma image classification,” *European Journal of Cancer*, vol. 119, pp. 11–17, 2019.
6. X. Liu, J. Faes, A. Kale, et al., “A comparison of deep learning performance against healthcare professionals in detecting diseases from medical imaging,” *The Lancet Digital Health*, vol. 1, no. 6, pp. e271–e297, 2019.
7. Y. LeCun, Y. Bengio, G. Hinton, “Deep learning,” *Nature*, vol. 521, pp. 436–444, 2015.
8. I. Goodfellow, Y. Bengio, A. Courville, *Deep Learning*, MIT Press, 2016.
9. J. Devlin, M. Chang, K. Lee, K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” *NAACL*, 2019.
10. R. Al-Rfou, et al., “Speech recognition with deep neural networks,” *IEEE Signal Processing Magazine*, vol. 29, no. 6, pp. 82–97, 2012.
11. Y. Ganin and V. Lempitsky, “Unsupervised domain adaptation by backpropagation,” *ICML*, pp. 1180–1189, 2015.
12. E. Tzeng, J. Hoffman, K. Saenko, T. Darrell, “Adversarial discriminative domain adaptation,” *CVPR*, 2017.
13. S. Daneshjou, et al., “Disparities in dermatology AI performance across skin tones,” *Nature Medicine*, 2021.
14. M. Groh, C. Harris, R. Soenksen, et al., “Evaluating deep neural networks trained on clinical images in dermatology,” *Nature Medicine*, 2021.
15. Z. Zhang, B. Lemoine, M. Mitchell, “Mitigating unwanted biases with adversarial learning,” *AAAI*, 2018.
16. D. Madras, E. Creager, T. Pitassi, R. Zemel, “Learning adversarially fair and transferable representations,” *ICML*, 2018.
17. J. Pearl, *Causality: Models, Reasoning, and Inference*, Cambridge University Press, 2009.
18. S. Ji, W. Xu, M. Yang, K. Yu, “3D convolutional neural networks for human action recognition,” *IEEE TPAMI*, 2013.
19. ISIC Archive, “International Skin Imaging Collaboration Dataset,” [Online]. Available: <https://www.isic-archive.com>
20. [ DermNet NZ, “Skin Disease Image Database,” [Online]. Available: <https://dermnetnz.org>